Google Summer of Code 2026
Machine Learning for Science (ML4SCI)

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Google Summer of Code @ ML4SCI Machine Learning for Science

GSoC 2026

Project Proposal

By

Sourish Dutta

On

Validating Neural Synchrony Patterns in Human Conversation (NeuroDyads)

Brain-to-Brain Decoder: Validating Neural Synchrony Patterns in Human Conversation

Sourish Kumar Dutta
B.Tech | IIT (ISM) Dhanbad, India
duttasourish12@gmail.com | github.com/sourishkumardutta
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Project	NEURODYAD — Brain-to-Brain Decoder
Organization	ML4SCI / University of Alabama
Mentors	Dr. Evie Malaia (Univ. of Alabama) & Dr. Brendan Ames (Univ. of Southampton)
Duration	350 hours, 12 weeks (Full-time)
Difficulty	Intermediate
Contact	ml4-sci@cern.ch

Abstract

I'm excited to share my plan for contributing to the NeuroDyads project this summer. Building on the great work started some years ago, my goal is to develop a machine learning pipeline for hyperscanning EEG data that isn't just powerful, but also rigorous and explainable.

I've broken my focus down into three main goals:

- **Reliable Testing**: I will build a stratified cross-validation pipeline specifically designed for dyadic EEG datasets to ensure our results are consistent.

- **Meaningful Metrics**: I plan to implement cross-entropy distance metrics, which will allow us to accurately compare how CEBRA embedding distributions differ across various neurotype groups.

- **Clear Insights**: I'll be conducting explainable analyses to map specific embedding features back to actual brain regions and frequency bands and also very valuable neurological observations.

I'm ready to hit the start working on the real project from day one onwards. I've already completed a full end-to-end preprocessing and CEBRA embedding pipeline for my pre-task, which achieved a **98.35% KNN decoding accuracy**. I'm eager to bring that same momentum to the rest of the project!

About Me

I am a B.Tech student in Petroleum Engineering at IIT (ISM) Dhanbad. I have also chosen to study computer science and machine learning. I am interested in representation learning, neural signal analysis, and artificial intelligence for scientific discovery. I have been deepening my knowledge in these areas outside my regular courses.

My technical skills include proficiency in PyTorch, NumPy, scikit-learn, MNE-Python, and CEBRA. My personal projects involve building neural networks from scratch. I have created a scalar version of PyTorch's Autograd, participated in Kaggle competitions, and earned certifications in Machine Learning from Stanford/Coursera and CNNs from DeepLearning.AI. I am a proud member of CyberLabs, IIT ISM's main machine learning research club. I also know C, C++, HTML, CSS, and JavaScript.

I find the NeuroDyads project really intriguing because it examines brain dynamics during two-person interactions. This focus is especially important in autism research. Discovering whether two brains truly coordinate during these interactions, and identifying which brain signals are key to this coordination, would be a major advancement in neuroscience. I would love to be part of this work.

Why NeuroDyads and ML4SCI

Most ML research focuses on a single area. NeuroDyads operates at the intersection of contrastive representation learning, clinical neuroscience, and social cognition, asking important open questions. The 2025 GSoC Project (Glushanina et al., arXiv 2025) showed that CEBRA can decode speaker-listener roles and demographic features from joint EEG data with high accuracy. However, the mechanism is still unclear: which brain regions are involved? Which frequency bands are important? How do embedding structures differ across dyads with varying neurotypes?

These are the questions the 2026 project aims to address, and they reflect the kind of rigorous, explainability-focused challenges I want to tackle. The finding by Crompton et al. (2025, Nature Human Behaviour) shows that communication issues arise from neurotype mismatch rather than individual shortcomings, making this work clinically relevant.

I also chose ML4SCI because the organization has a strong history of producing research-quality outputs through GSoC. Last year's NeuroDyads preprint is a clear example of that. I want to contribute to work that gets published on arXiv, not just a summer project.

Pre-Task: What I Built and What I Learned

For my pre-task, I built a complete pipeline that takes raw EDF files and turns them into interpreted CEBRA embeddings. It was a great way for me to get under the hood of the NeuroDyads workflow.

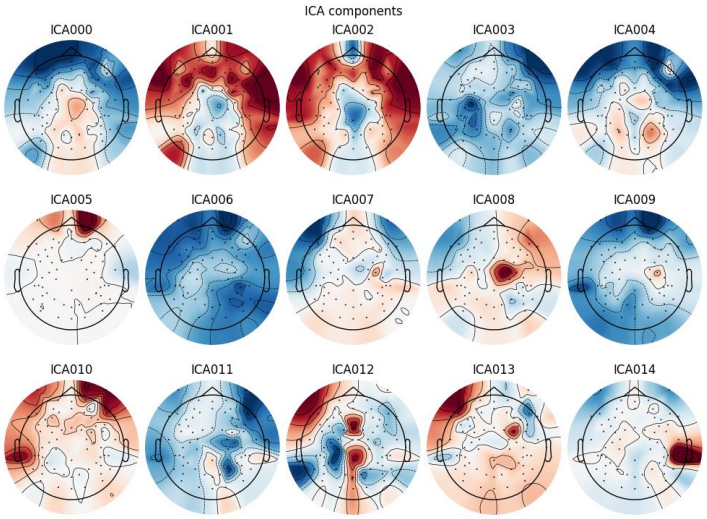
Here is a quick look at the technical choices I made and why I believe they are the right fit for our 2026 goals:

Part 1 - Preprocessing

I loaded two raw 65-channel EDF files (Speaker: NT#10, Listener: NT#9) at 250 Hz. I verified the zero-offset DIN1 markers. I applied a 0.1 to 45 Hz bandpass filter using a Hamming FIR filter. I dropped the VREF reference channel and renamed the channels to match GSN-HydroCel-65_1.0 standard labels. I also set the electrode montage for ICA compatibility.

For bad channel detection using PyPrep (RANSAC), I identified 10 bad channels in the Speaker file (E7, E21, E23, E27, E29, E31, E32, E52, E62, E63) and 6 in the Listener file (E10, E23, E41, E55, E62, E63). These were found through deviation, HF noise, correlation, and RANSAC criteria. I interpolated them using spherical splines. This step is essential because CEBRA assumes the joint T by 128 matrix is meaningful at every time step. A noisy or flat channel affects every sample it participates in.

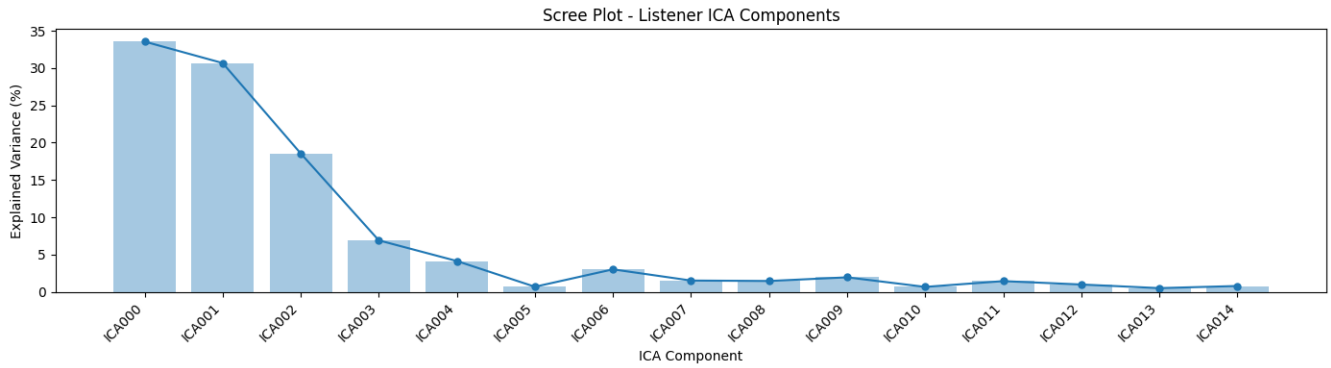
For **ICA artifact removal**, I chose Extended Infomax ICA for both participants. I used 20 components for the Speaker and 15 for the Listener. The number of components was justified using the data-to-component rule, which requires $30N^2$ data points.



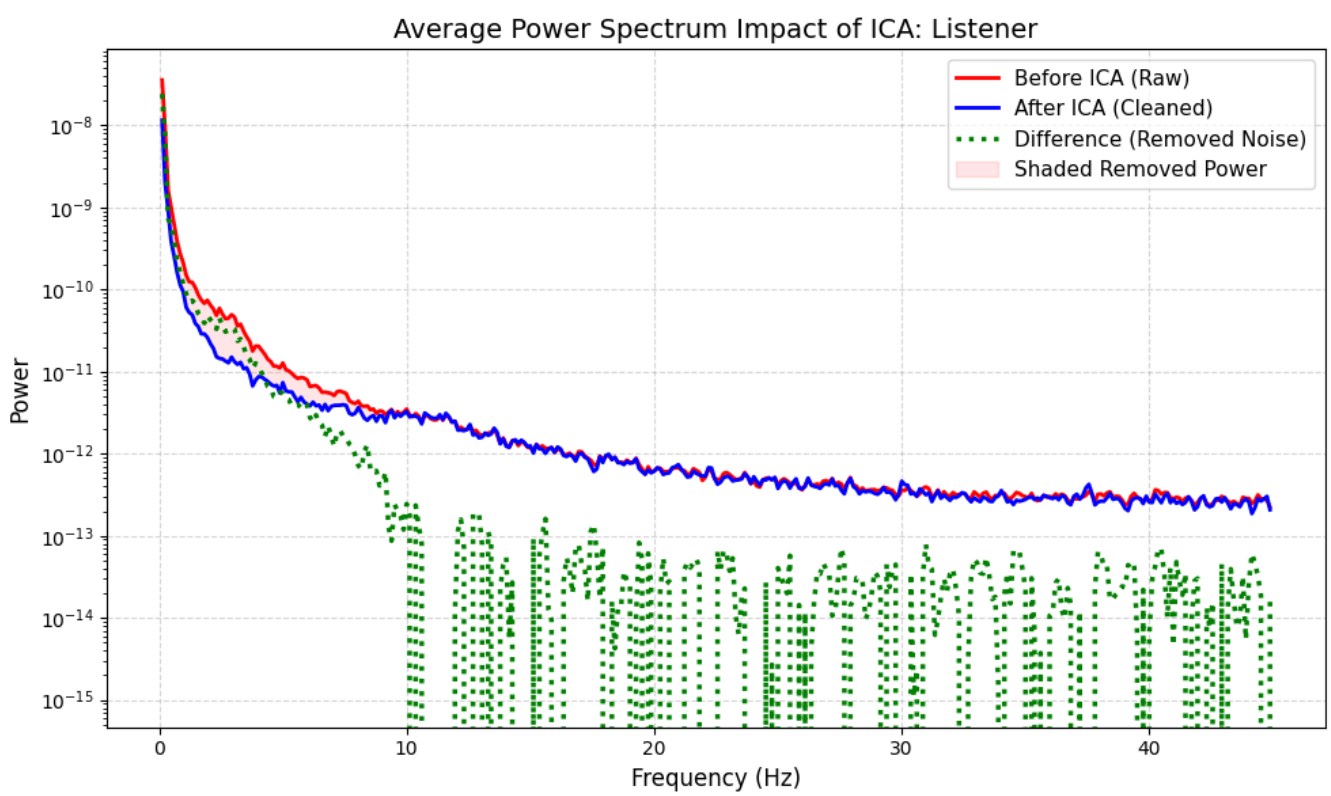
The 20 components used about 16% of the Speaker's 75,411 samples, which is within the reliable range. I excluded 8 components from the Speaker (eyeblinks: ICA003, ICA004, ICA008; saccades: ICA006; jaw/muscle: ICA001, ICA009, ICA015; electrode artifact: ICA019) and 4 from the Listener (eye movements: ICA001, ICA002, ICA004; cardiac/eye: ICA006). I made sure to run ICA separately for each participant. The image on the right shows the ICA of the listener dataset after processing and eliminating certain components.

Joining two different brains into a single ICA decomposition would violate the assumption of statistical independence and create unwanted mixing between participants. I documented this decision in the notebook.

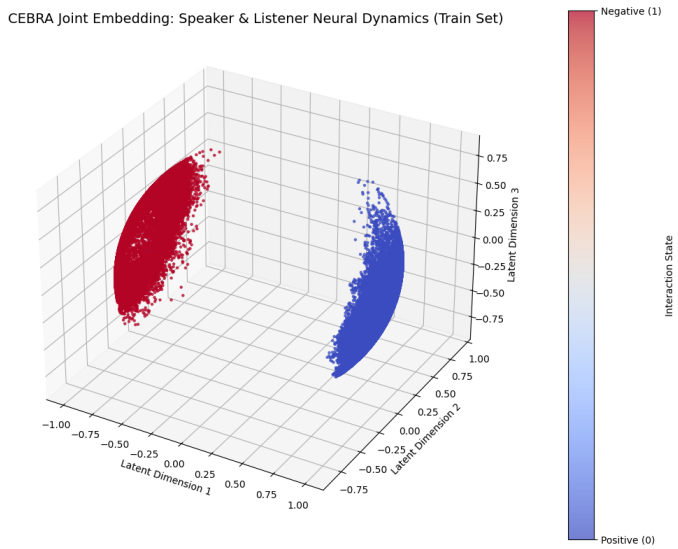
One **observation** I made was that after ICA, the back-projected signal showed slightly higher high-frequency power due to rank deficiency from bad channel interpolation. This is a known mathematical artifact of ICA reconstruction on rank-reduced data, not an issue with signal quality. I noted this reasoning in the notebook.

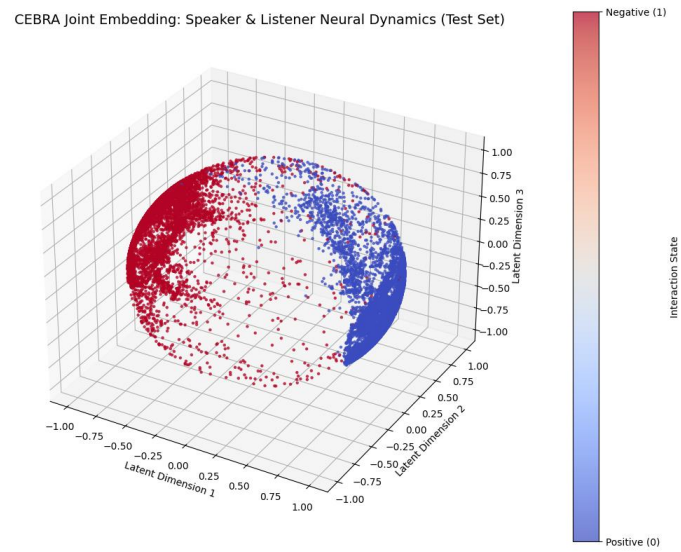


Improvements over the 2025 baseline: The original NeuroDyads pipeline used 15-component ICA with a fixed exclusion array. While reviewing the topographies, I found additional ocular artifacts (ICA002, ICA004 — distinct vertical blinks) and a saccade component (ICA006) that were not excluded in the baseline. I removed these additional components from the Speaker data to lessen CEBRA's tendency to learn from large-variance eye movements instead of true neural activity.



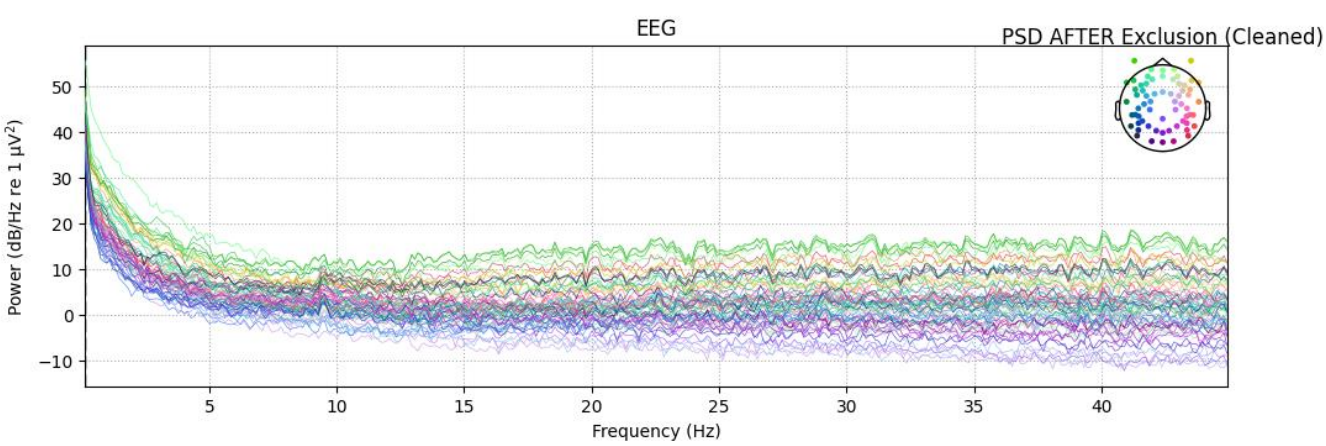
Part 2 - CEBRA Embedding





I combined the preprocessed Speaker and Listener data into a T×128 matrix. The first 64 columns represent the Speaker, while the last 64 columns represent the Listener. Then, I z-normalized each channel independently over time and trained CEBRA with affective labels (0 for positive, 1 for negative) using an 80/20 train-test split that maintains the temporal structure to avoid leakage.

CEBRA configuration: offset10-model, cosine distance, time_delta conditioning, 3D output, 5000 iterations, batch size 512, learning rate 3e-4. This approach closely follows the 2025 pipeline (Glushanina et al.) but uses time_offsets=5 instead of 10 to accommodate the shorter segment length of this single-dyad dataset.

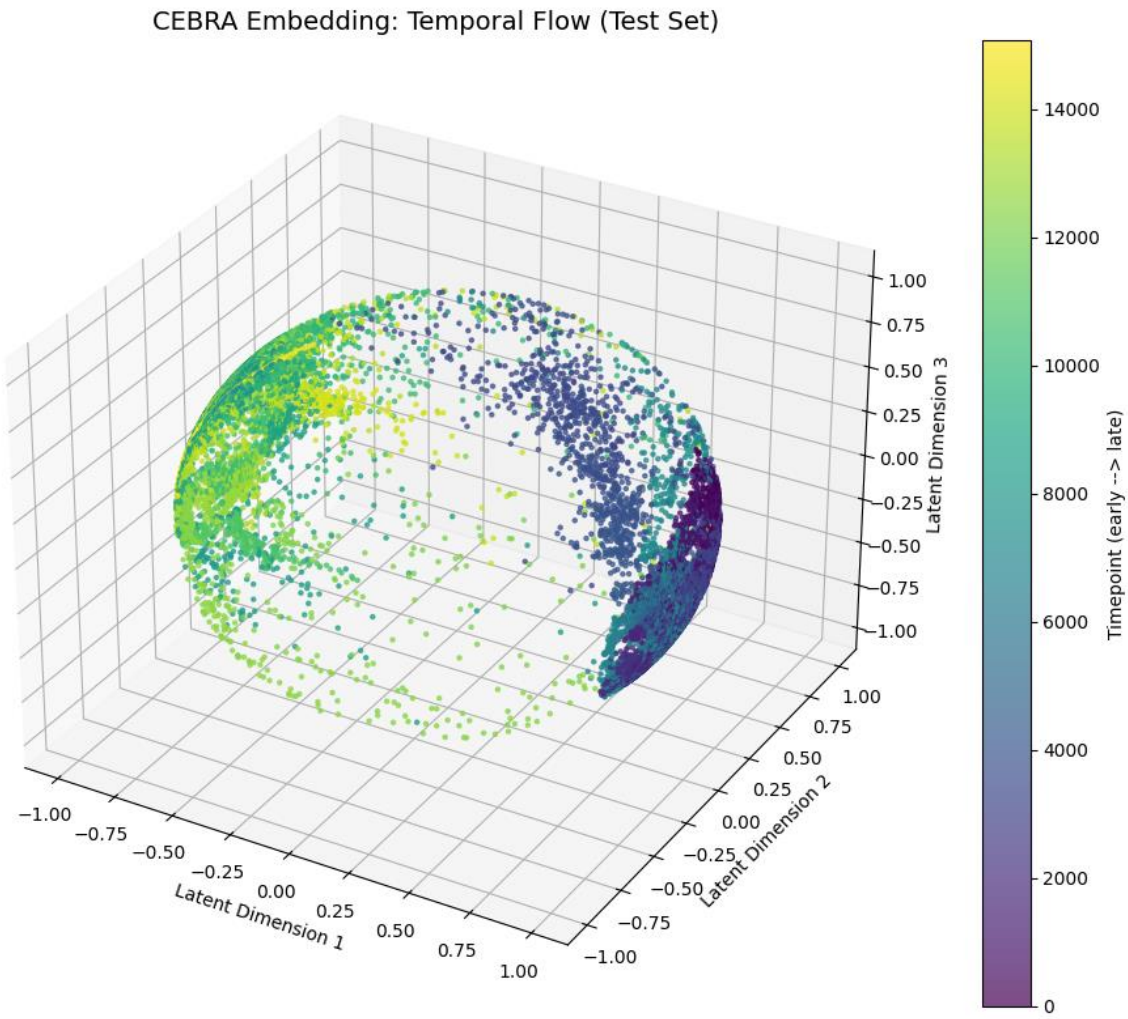


Results of the Main model:

KNN-5 decoding accuracy: 98.35% | InfoNCE Loss (GoF): 5.74

Control model (shuffled labels): 49.45% | GoF: 6.17

The accuracy of the control model dropped to **nearly chance levels (49.45%).** This confirms that the main model's structure relies on the labels, rather than just on temporal autocorrelation.



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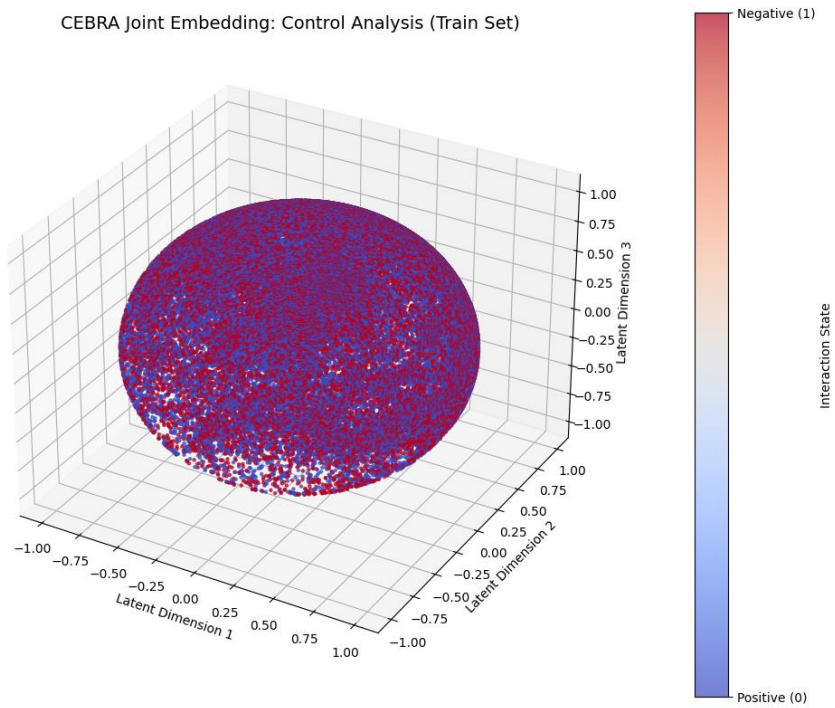
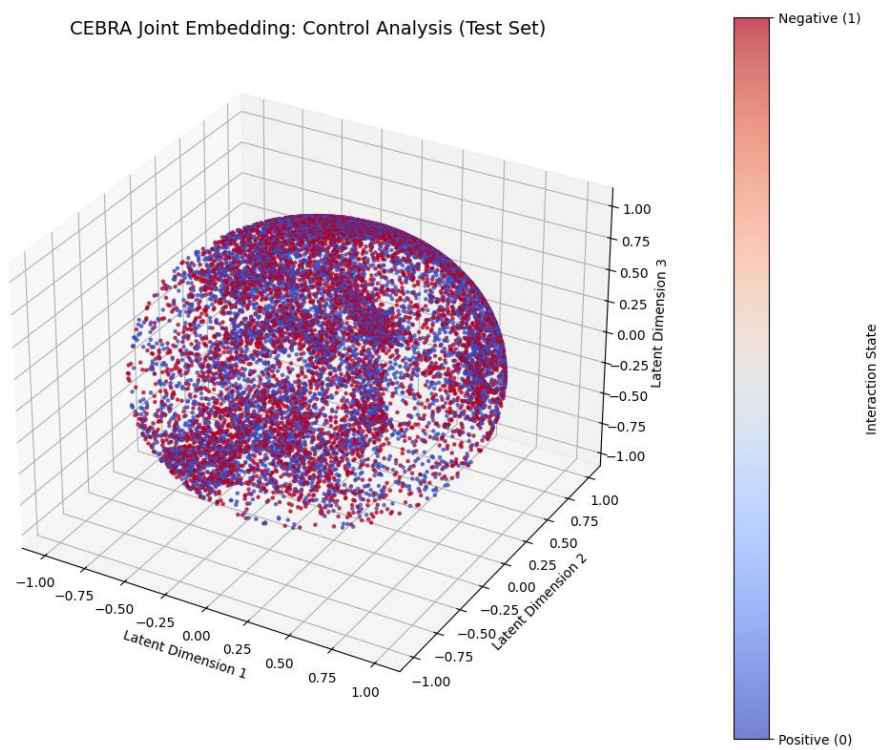
Part 3 - Embedding Geometry

In Part 3, I noticed two clear 3D clusters for positive and negative feelings, with a thin transition area. This represents the neural boundary between different conversation states. Temporal coloring revealed sub-clusters within each condition, showing changes in the joint brain state from moment to moment.





Part 4 — Critical Reflection

The last part, which is the most important section, identifies the main limitation of the model due to my specific choices of the pipeline and ICA removal. The binary affect gives CEBRA almost no variability within conditions to learn from. The model likely uses temporal position as a stand-in for the label since both change together perfectly. A proper cross-shift control, which offsets one participant's stream by about 30 seconds, would clarify whether CEBRA learned about dyadic co-occurrence or just individual temporal patterns. I see this missing control as the most critical addition I would make if I had more time.

I also found that performing independent ICA for each participant disrupts the synchronized artifact rejection method used in the 2025 paper, which removes identical time segments from both streams at the same time. This is a preprocessing alignment issue I plan to tackle systematically in the GSoC pipeline. I used this particular methodology here as the EEG data I had was less and removing segments from it could result in an unbalanced dataset.

Please refer to the IPYNB Notebook in uploaded in my github (link given in the beginning of this proposal) for the detailed Part 4 Reflection.

Project Plan

The 2026 NeuroDyads project scales from 8 to 40+ dyads and shifts focus from regular baseline decoding to rigorous validation and explainability.

My 12-week plan is organized around three deliverables, each building on the last.

Deliverable 1 - Stratified Cross-Validation Pipeline (Weeks 1–4)

The central methodological challenge with dyadic EEG data is that standard cross-validation leaks information: if trials from the same dyad appear in both train and test sets, the model can overfit to pair-specific signatures rather than learning generalizable patterns. I will build a dyad-aware stratified cross-validation framework that treats each dyad as an indivisible unit.

- **Implement leave-one-dyad-out (LODO**) CV as the primary evaluation scheme, consistent with the N+1 analysis in Glushanina et al. (2025)

- **Implement stratified k-fold** at the dyad level, ensuring proportional representation of neurotype combinations (autistic-autistic, autistic-neurotypical, neurotypical-neurotypical) in each fold.

- **Implement the cross-shift control:** offset Participant B's time series by a fixed lag (~30 s) and retrain CEBRA to test whether dyadic co-occurrence — not individual temporal trajectories — drives decoding accuracy

- **Generate learning curves:** KNN accuracy and GoF as a function of number of dyads, to determine the sample size at which neural decoding stabilises and justify the 40+ dyad target

- **Reproduce the pseudocut check** from the 2025 paper on the full dataset to confirm that preprocessing cuts do not introduce embedding artifacts.

Deliverable 2 - Cross-Entropy Distance Metrics (Weeks 5–8)

The Roca et al. (2023, Cell Reports Methods) cross-entropy test offers a clear, model-free method to compare embedding distributions between groups. Instead of looking at summary statistics like mean accuracy or GoF, cross-entropy distance shows us if the geometric structure **of two embeddings** is identical. In other words, it indicates whether the same neural dynamics are involved.

- **Implement the cross-entropy distance pipeline** from Roca et al. 2023: train a reference CEBRA model on one group, compute the encoding of a second group's data, and measure the distributional distance between embeddings

- **Apply to three group comparisons**:
  a) same-neurotype (autistic-autistic vs. neurotypical-neurotypical)
  b) mixed vs. unmixed dyads
  c) same-gender vs. cross-gender dyads (replicating the gender-related finding from the 2025 N+1 analysis)

- **Compute confidence intervals** via bootstrap resampling and assess statistical significance
- **Visualize the distance matrix** across dyad groups and annotate with AQ score differences to assess whether neurotype mismatch continuously predicts embedding distance

Deliverable 3 → Explainability: Feature-to-Brain Mapping (Weeks 9–11)

The 2025 ablation analysis (removing channels associated with speech production or executive function) found no significant impact on decoding accuracy, suggesting distributed neural representations. I will go further! Rather than removing channels and checking whether performance degrades, I will identify which channels and frequency bands positively contribute to the learned embedding.

- **Implement channel contribution analysis**: for each of the 128 channels (64 per participant), compute the change in InfoNCE loss when that channel's time series is replaced with Gaussian noise. Rank channels by contribution magnitude and visualize as a topographic map

- **Implement frequency band decomposition:** bandpass-filter the raw EEG into standard bands (delta 1–4 Hz, theta 4–8 Hz, alpha 8–12 Hz, beta 12–30 Hz, gamma 30–45 Hz) and train separate CEBRA models per band to determine which frequencies drive decoding

- **Combine channel and frequency analyses**: identify which (electrode, frequency) pairs most inform classification, and compare spatial maps between speaker and listener to test for hemispheric or regional asymmetries in inter-brain synchrony

- **Relate the top contributing features** to known neurophysiology (e.g., theta-band prefrontal synchrony in social cognition, alpha-band temporal regions in speech comprehension)

Buffer / Documentation Week (Week 12) :

- **Finalize all code**, write comprehensive docstrings, and integrate the pipeline into the NeuroDyads GitHub repository
- Prepare a **brief technical write-up** summarizing methods and key findings, suitable for inclusion in the project's forthcoming publication
- **Address any feedback from mentors, resolve open issues, and ensure reproducibility** on the full 40+ dyad dataset

Timeline at a Glance

Phase	Dates	Deliverables
Community Bonding (May)	May 8 – June 1	Deep-read Roca et al. 2023 and full NeuroDyads codebase. Set up environment. Discuss prioritization with mentors.
Weeks 1–2	June 2–15	Dyad-aware LODO and stratified k-fold CV framework. Learning curve generation.
Weeks 3–4	June 16–29	Cross-shift control. Pseudocut check on full dataset. Preliminary cross-dyad results.
Weeks 5–6	June 30–July 13	Cross-entropy distance pipeline (Roca et al.). Group comparison: neurotype and gender.
Weeks 7–8	July 14–27	Bootstrap confidence intervals. Distance matrix visualization. Midterm evaluation.
Weeks 9–10	July 28–Aug 10	Channel contribution (noise masking) analysis. Topographic maps.
Week 11	Aug 11–17	Frequency band decomposition. Combined (channel, band) explainability maps.
Week 12	Aug 18–25	Integration, documentation, write-up, mentor review.

End-Term Evaluation & Wrap-up:

By the end of the program, I will ensure all implementations are thoroughly documented, tested, and accessible in the project repository. I aim to provide clear examples and instructions for reproducing the results.

I plan to actively communicate progress weekly with my mentors and incorporate their feedback throughout the project. The buffer week is included to manage any potential blockers effectively.

Following GSoC, I am interested in contributing further, potentially through blog posts explaining the work or contributing to any resulting research papers.

Technical Skills & Relevant Experience

Directly Relevant

- MNE-Python: Full preprocessing pipeline (filtering, bad channel detection, ICA, segmentation) demonstrated in the NeuroDyads pre-task
- CEBRA: Trained and evaluated contrastive embedding models, including control analyses and GoF metrics, on real hyperscanning EEG data
- PyPrep: RANSAC-based automated bad channel detection
- NumPy, SciPy, scikit-learn: Data manipulation, z-normalization, KNN classification, train-test splits
- PyTorch: Built autograd engines and neural networks from scratch (Micrograd project, validated against PyTorch autograd)

Supporting

- ML competition experience: Finished 3rd on Kaggle private leaderboard (CyberLabs WOC); demonstrated OOF cross-validation, class imbalance handling, and probability calibration
- 1st place in PhysisTechniques Hackathon – Built QSetum, an optical setup generator based on given quantum targets.
- 2nd place, AI Summit 2026 (among 60–80 teams) - autonomous job-search agent using FastAPI and Gemini API
- Certifications: ML Specialization (Stanford/Coursera, Dec 2025), CNNs (DeepLearning.AI, Mar 2026)
- Active member, CyberLabs ML Division, IIT (ISM) Dhanbad (Jan 2026 - present)

Commitment and Availability

GSoC 2026 will be my sole major commitment during the coding period. I have no summer internship, coursework conflicts, or competing obligations. I can commit 35–40 hours per week throughout the 12-week period.

I am in IST (UTC+5:30). I am comfortable communicating via the project mailing list (ml4-sci@cern.ch) and also via video call at times convenient for mentors in the US and UK. I will share weekly progress updates and push code to GitHub on a regular cadence.

I have already engaged deeply with the project: I have read the NeuroDyads preprint (arXiv 2509.23479v1), the CEBRA paper (Schneider et al., Nature 2023), the Roca et al. cross-entropy paper (Cell Reports Methods 2023), and the Crompton et al. communication paper (Nature Human Behaviour 2025) referenced in the project description. I checked the original NeuroDyads GitHub repository. I used it to explain my preprocessing choices in the pre-task.

The Motivation for GSoC

I am deeply enthusiastic about programming, problem-solving, and the power of machine learning. Through my involvement in the Cyberlabs ML division, I've developed a strong interest in how models learn complex patterns and how ML techniques can be applied to scientific and real-world challenges. Contributing to open-source projects is incredibly valuable to me – it's a chance to collaborate with diverse communities, sharpen my technical abilities, and build software with tangible impact. Google Summer of Code represents an unparalleled opportunity to channel my passion, learn directly from experienced mentors in the open-source world. I'm constantly motivated to expand my knowledge by delving into research papers and staying updated with the latest advancements in technology and making a meaningful contribution to impactful technology.

Conclusion

The **NeuroDyads** project tackles a difficult issue: understanding how two brains work together during natural conversations and what this interaction reveals about typical and atypical social communication. The 2025 GSoC pilot demonstrated that it is feasible.

The 2026 project poses a more challenging question: why does it work? It explores which features are important, how reliable the results are, and how different structures compare across groups.

I have done more than just read about this project. I built the pipeline and made my own data-driven choices at every stage. I've identified its true limitations and documented my reasoning thoroughly. My pre-task submission, which achieved 98.35% KNN accuracy on a well-cleaned dyadic EEG dataset, includes a thorough control analysis and an honest critical reflection. This shows that I can make meaningful research contributions, not just follow tutorials.

I would be honored to work with Dr. Malaia and Dr. Ames to advance this project.

Thank You